



Airbnb App Plan

Group 1 - Adam, Roisin, Tariq,
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Agenda

	Scope
	Team Roles
	Project Set-Up
	Wireframe Mockup
	Enrichment Datasets
	Investment Scoring
	Investor Profiles and Stakeholders
	Application Plan
	Project Timeline

Scope

Objective

- Help property investors make informed property investment decisions
- Analyse short-term Airbnb and long-term rental opportunities across four UK cities
- Generate data-driven recommendations for different investor profiles

Key outputs

- Top neighbourhoods
- Top property types
- Investment score
- Risk assessment
- AI-generated summaries

Roles

Engineer 1 - Adam

Engineer 2 - Tariq (Project Lead)

Analyst 1 - Roisin

Analyst 2 - Thadsha

Analyst 3 - Josh

Team Structure

Person	Role	Week 1-2 Responsibilities
Data Engineer 1	Data Acquisition Lead	Find datasets, download Airbnb data, investigate other data sources (House Prices, Crime, ONS, Land Registry)
Data Engineer 2	Architecture Lead	Design folder structure, GitHub repo, data pipeline plan, app architecture, deployment research
Data Analyst 1	Business & Personas Lead	Define investor personas, user stories, requirements, success criteria
Data Analyst 2	Analysis Lead	Design investment score methodology, KPIs, metrics, assumptions
Data Analyst 3	Dashboard & AI Lead	Wireframes, app pages, AI feature planning, recommendation logic

Project Set-Up



Trello board
created



GitHub repository
established



Daily stand-up
meetings



Databricks/Python
environment
setup



Task allocation
completed

Logo



Wireframe

Data-Driven Real Estate Investment

Make informed decisions with powerful analytics and insights

Get Started



Persona Selection

Select your investment persona



Area Comparison

Analyze and contrast different neighborhood



Property Type Comparison

Compare performance across property types



Investment Scoring

Evaluate properties with comprehensive scores



Map Visualization

Explore investment opportunities on a map



Risk Dashboard

Monitor and manage Investment risks

Enrichment Datasets

Dataset	Purpose	Granularity	Source
Short-Term Rental Data	Compare Airbnb vs long-term rental returns	<i>Coordinates</i>	<i>insideairbnb.com</i>
Median House Price Data	Estimate investment cost and long-term ROI	<i>Middle layer Super Output Area (MSOA)</i>	English Data: <i>ons.gov.uk</i> Scottish Data: <i>ros.gov.uk</i>
Crime Data	Assess neighbourhood safety	<i>Coordinates</i>	<i>data.police.uk</i>
Transport Data	Measure distance from railway stations and bus stops to assess accessibility and attractiveness	<i>Coordinates</i>	<i>NaPTAN (National Public Transport Access Node)</i>
Amenities Data	Number of GP surgeries, libraries, parks in area	<i>Local Authority District (LAD)</i>	<i>ons.gov.uk</i>

Scoring

- Develop a transparent scoring model to identify the most attractive Airbnb investment opportunities.

Factor	Key Metrics
Revenue Potential	Measures how much income a property could generate as an Airbnb.
Demand Indicators	Review Count, Review Recency, Availability
Guest Satisfaction	Review Score, AI Sentiment, Review Themes
Location Attractiveness	Transport Links, Amenities, Crime Rate
Financial Viability	House Price, Revenue-to-Price Ratio
Risk Assessment	Regulatory Risk, Market Saturation, Data Confidence

Investor Profiles

Lifestyle Property Fund

- A small angel syndicate focused on helping first-time buyers and existing homeowners maximise property income.

Urban Stay Partners

- A group of experienced Airbnb operators and small property management companies.

PropTech Capital

- A venture capital fund specialising in property technology and data-driven investment tools

Civic Insight Foundation

- A public-interest fund supporting responsible tourism, housing policy, and city planning.

Stakeholders

Homeowners & First-Time Buyers

- Compare Airbnb vs long-term rental income
- Need simple revenue estimates and risk guidance

Multi-Property Airbnb Hosts

- Optimise and expand Airbnb portfolios
- Need neighbourhood, pricing, and review insights

Property Investors & Developers

- Make data-driven investment decisions
- Need investment scores, yield comparisons, and market analysis

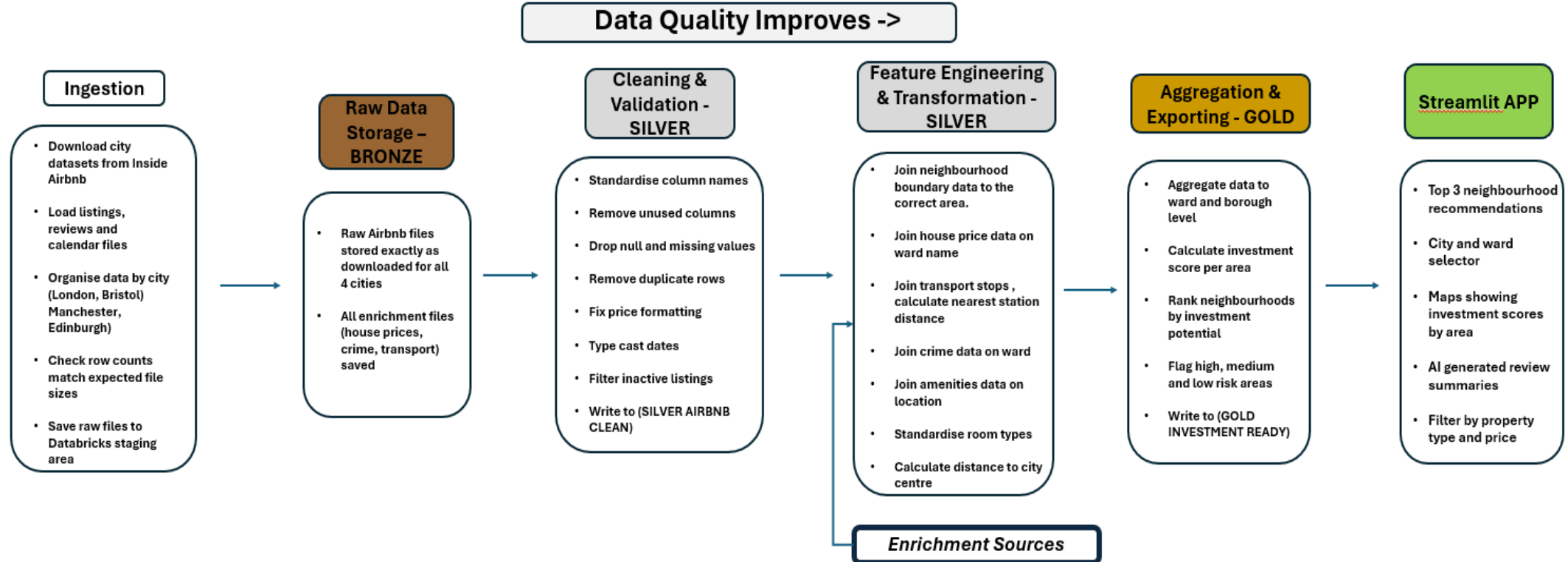
Policymakers & Tourism Boards

- Monitor Airbnb activity and tourism trends
- Need compliance, density, and neighbourhood-level insights

Application Plan

Airbnb Investment App Data Pipeline

Built on Databricks using Medallion
Architecture (Bronze / Silver / Gold)



Project Timeline

Phase	Status
Project Setup	✓ Complete
Data Collection	✓ Complete
Data Engineering	🔄 In Progress
Scoring Model	🕒 Upcoming
Dashboard Development	🕒 Upcoming
Final Testing	🕒 Upcoming

Any Questions?

Laws

Area	Airbnb Host	Landlord
Legal status	Short-term license	Tenancy
Guest rights	Very limited	Strong
Eviction	Immediate (trespass)	Court process
Safety	Fire + basic safety	Full housing safety
Tax	Rent-a-Room / trading	Property income
Planning	Restrictions in some areas ~ London 90-day rule	None
Insurance	Host guarantee + optional	Landlord insurance
Regulation	Light	Heavy